CS3230 Algorithm Analysis

"Math for AA"

- **Lecture Topics and Readings**
  - Simple Algorithm Analysis — CS1020/2010
  - Algorithm Analysis Pattern — [LHW] A&E
  - Asymptotic Notations — [CLRS]-C3
  - Functions & Summations (Review) — [CLRS]-C3, App

Quickly get up to speed.

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Software Design Patterns

Hon Wai Leong, NUS

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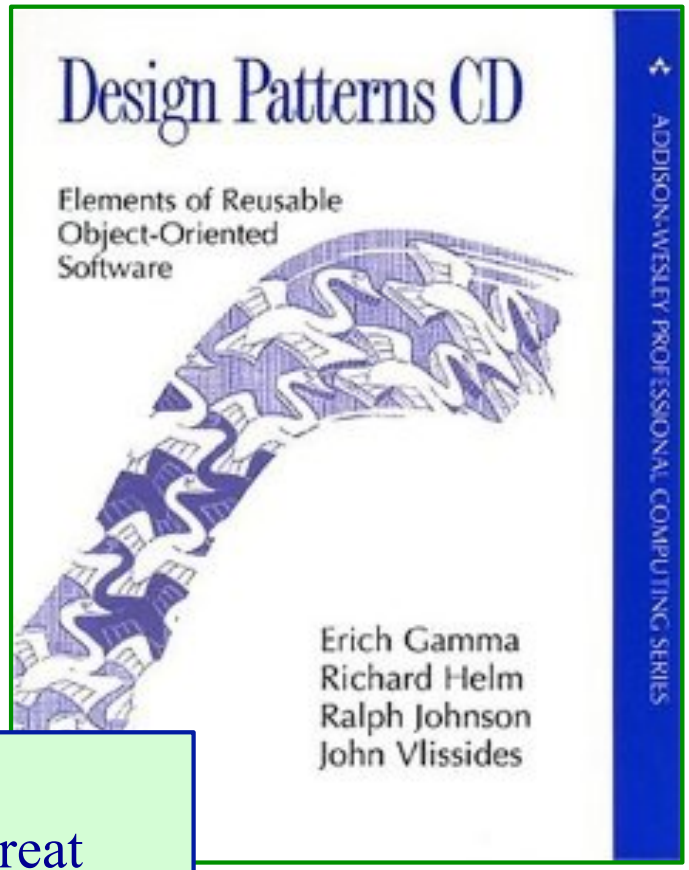
Software Design Patterns

- **Started with GoF book, 1995**
  - **Gave list of 23 patterns**

Design Patterns CD: Elements of Reusable Object-Oriented Software (Professional Computing) [CD-ROM]

Erich Gamma (Author), Richard Helm (Author), Ralph Johnson (Author), John Vlissides (Author)



"Published in 1995, Design Patterns: Elements of Reusable Object-Oriented Software has elicited a great deal of praise from the press and readers. The 23 patterns contained in the book have become an essential resource for anyone developing reusable software designs."

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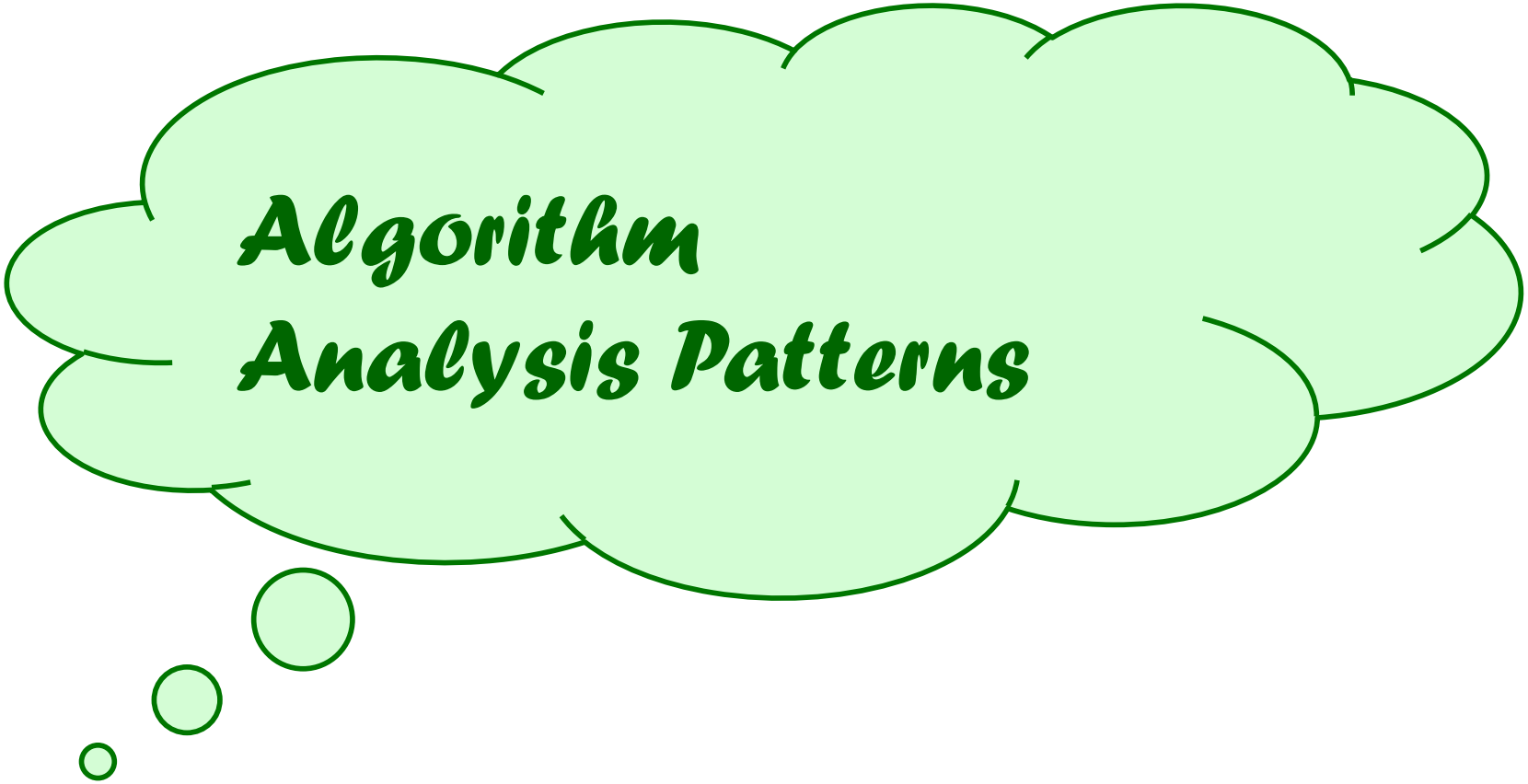
How about…



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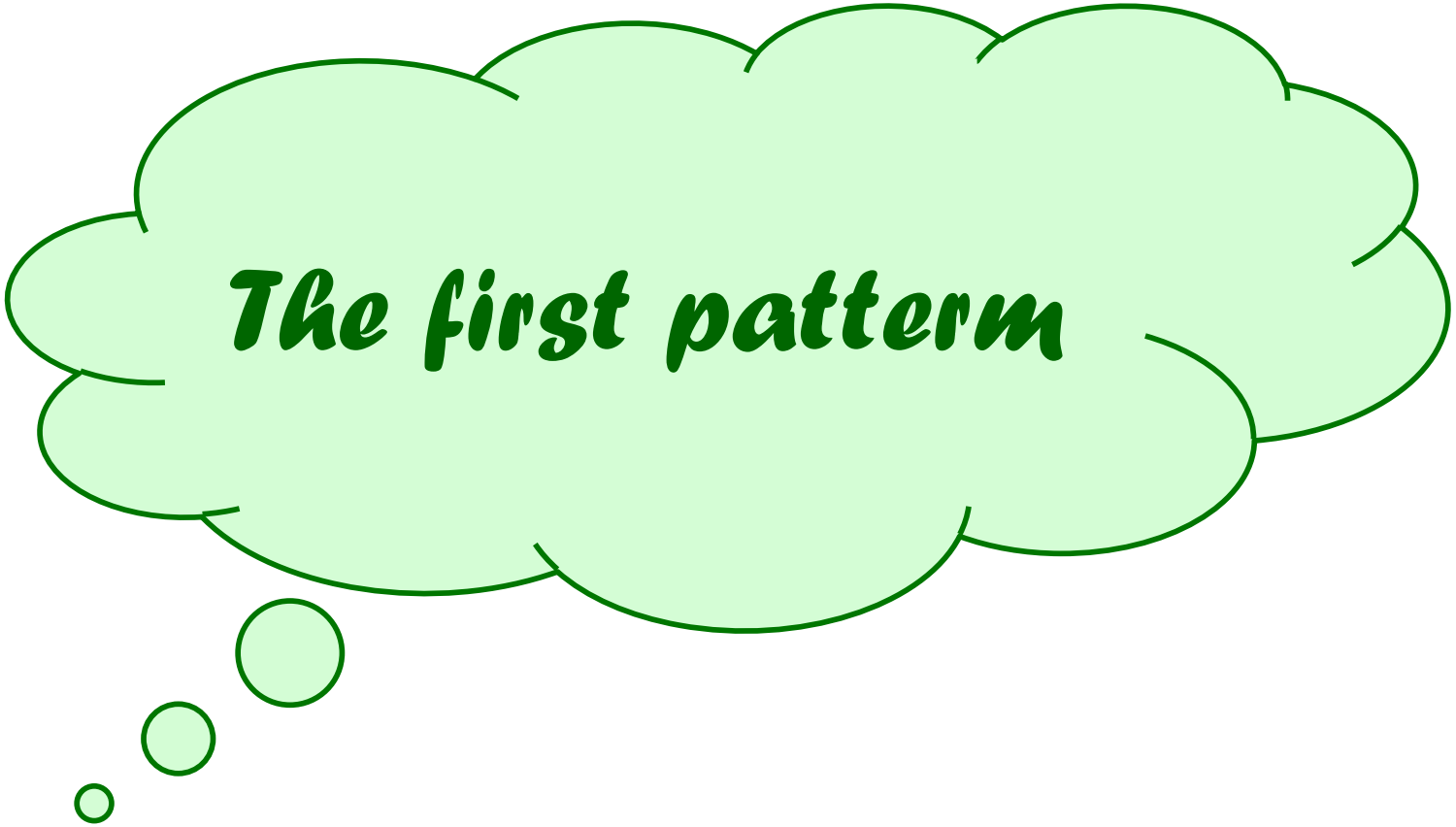
Algorithm-Analysis Pattern?



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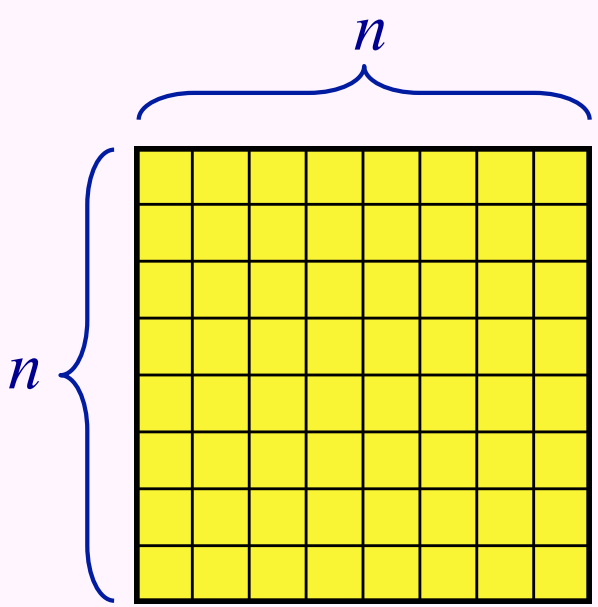
Algorithm Analysis Pattern

AA-Pattern: 2D-Box

$$T(n) = \sum_{k=1}^{n} cn = \sum_{k=1}^{n}\left(\sum_{j=1}^{n} c\right) = \Theta(n^2)$$



$T(n) = n^2$

Code Pattern

```
MATRIX-SUM (M, n, n) ▷ M[1..n; 1..n]
    Sumsf ← 0
    for k ← 1 to n
        for j ← 1 to n
            do Sumsf ← Sumsf + M[k, j]
    Sum ← Sumsf
```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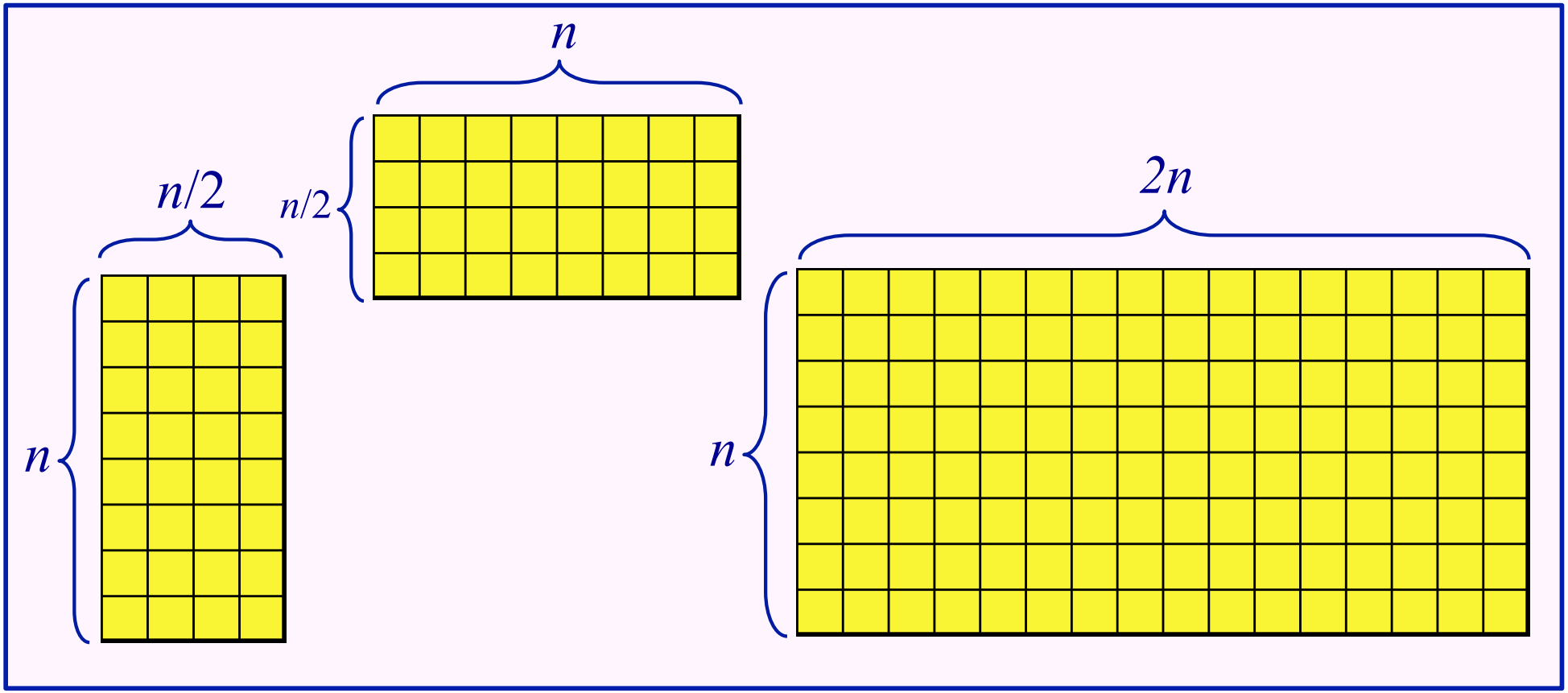
2D-Box Pattern (variants)

AA-Pattern: 2D-Box

$$T(n) = \sum_{k=1}^{dn} cn = \sum_{k=1}^{dn}\left(\sum_{j=1}^{n} c\right) = \Theta(n^2)$$



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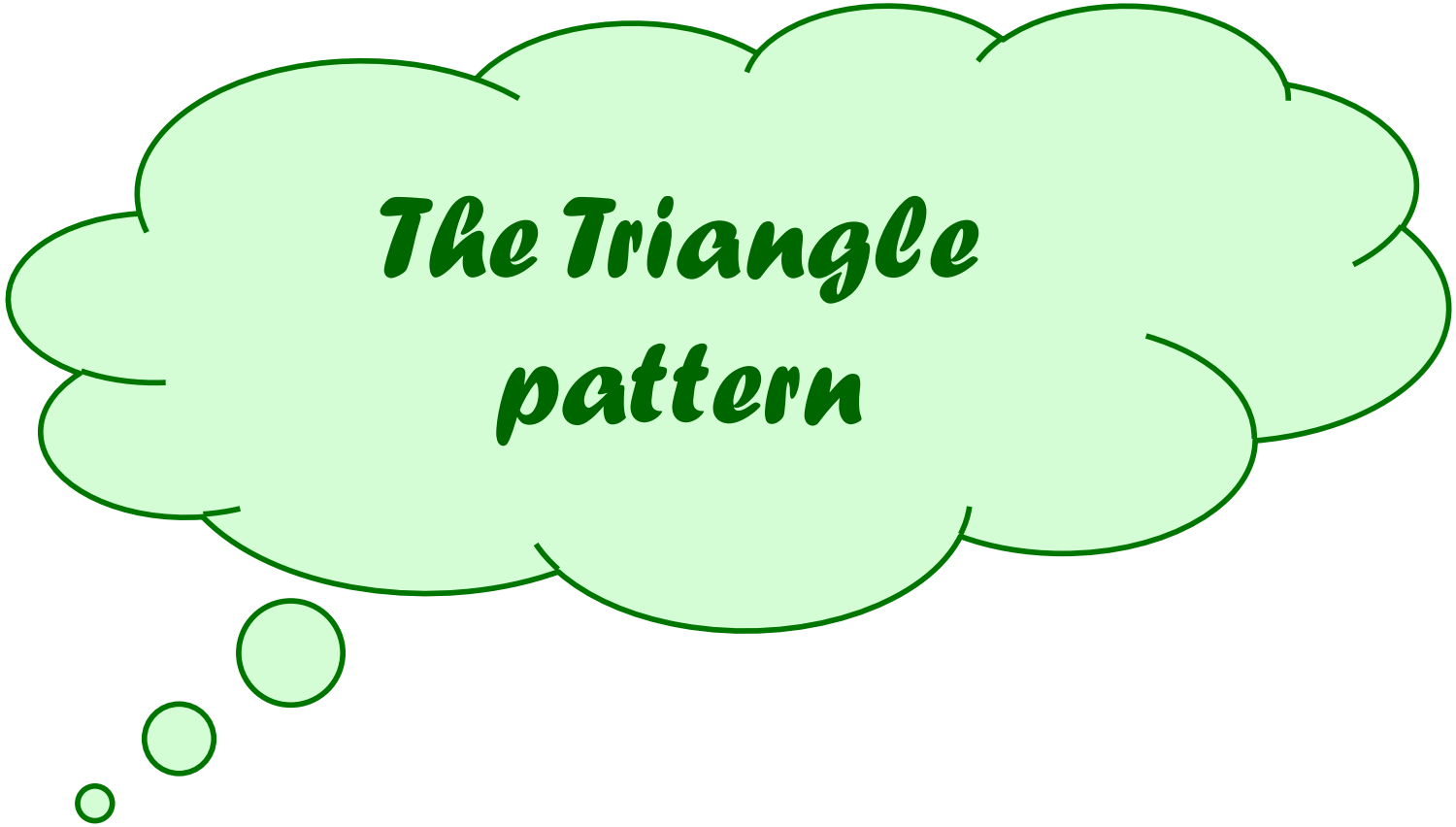
Algorithm-Analysis Pattern?



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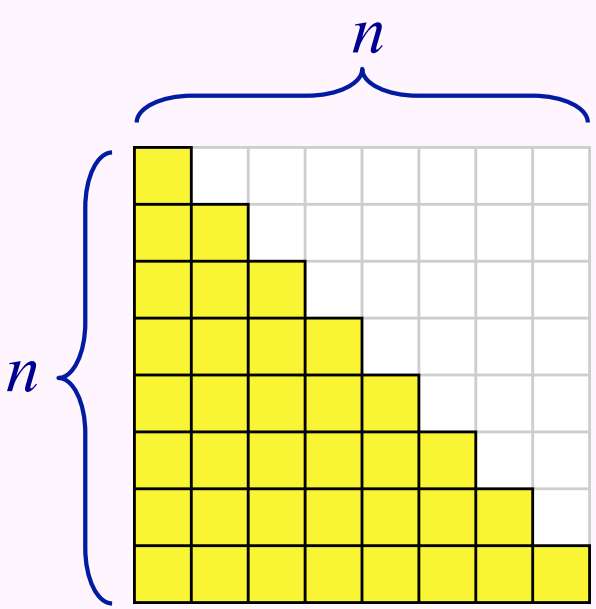
Algorithm Analysis Pattern

AA-Pattern: Triangle

$$T(n) = \sum_{k=1}^{n} ck = \sum_{k=1}^{n}\left(\sum_{j=1}^{k} c\right) = \Theta(n^2)$$



$T(n) = n(n+1)/2$

Code Pattern

```
TRIANGLE-SUM (M, n, n) ▷ M[1..n; 1..n]
    Sumsf ← 0
    for k ← 1 to n
        for j ← 1 to k
            do Sumsf ← Sumsf + M[k, j]
    Sum ← Sumsf
```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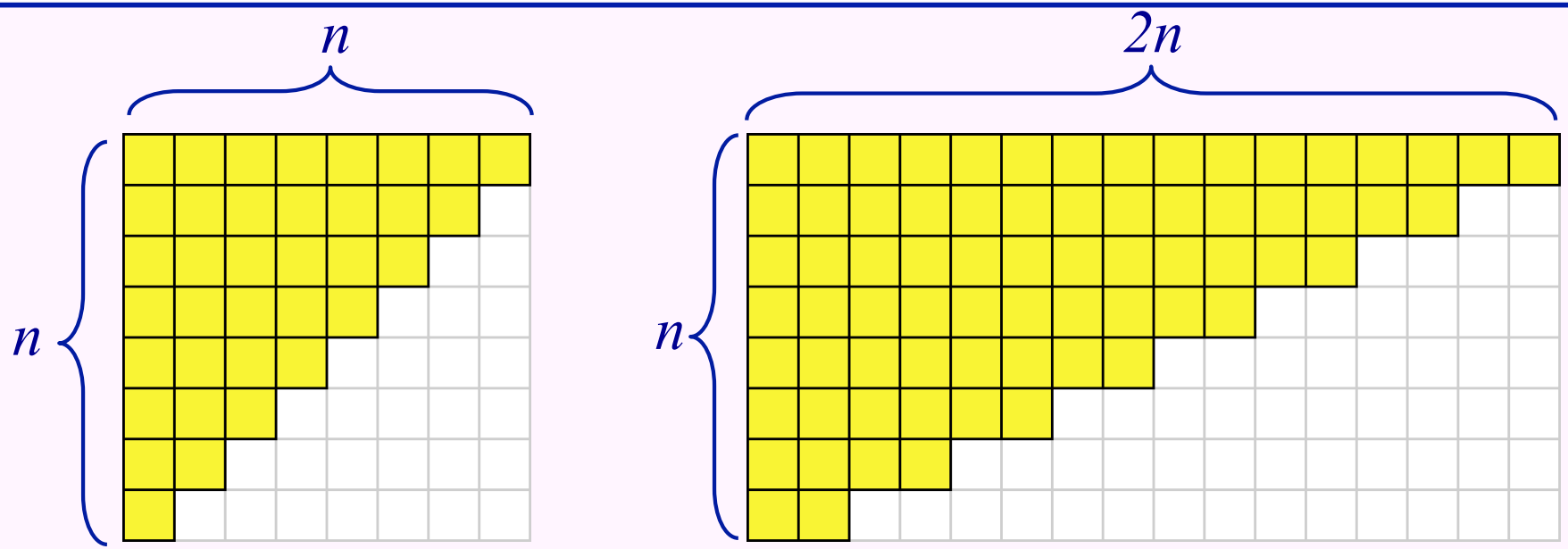
Triangle Pattern (variants)

AA-Pattern: Triangle

$$T(n) = \sum_{k=1}^{n} ck = \sum_{k=1}^{n}\left(\sum_{j=1}^{k} c\right) = \Theta(n^2)$$



$T(n) = n(n+1)/2$

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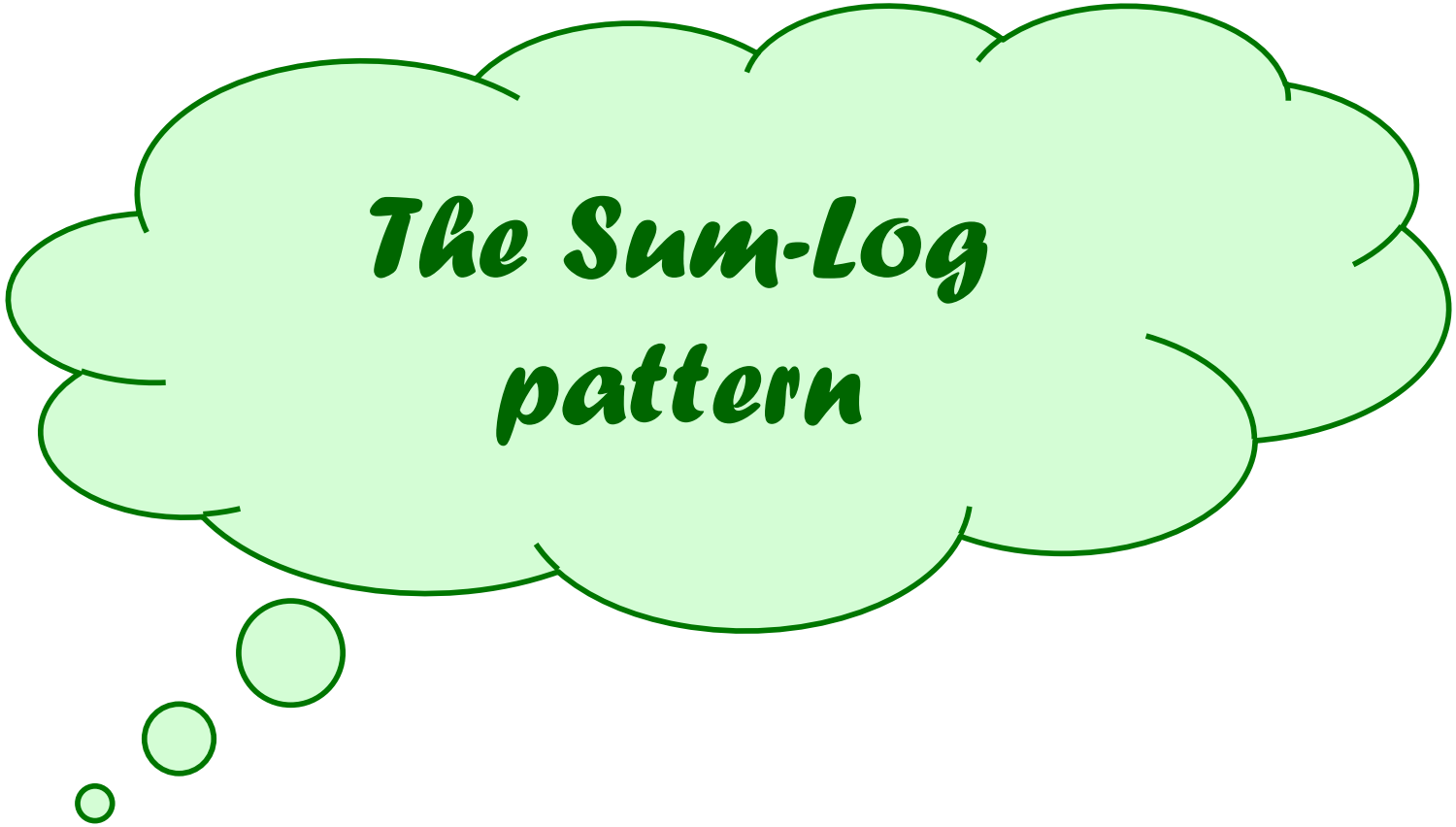
Algorithm-Analysis Pattern?



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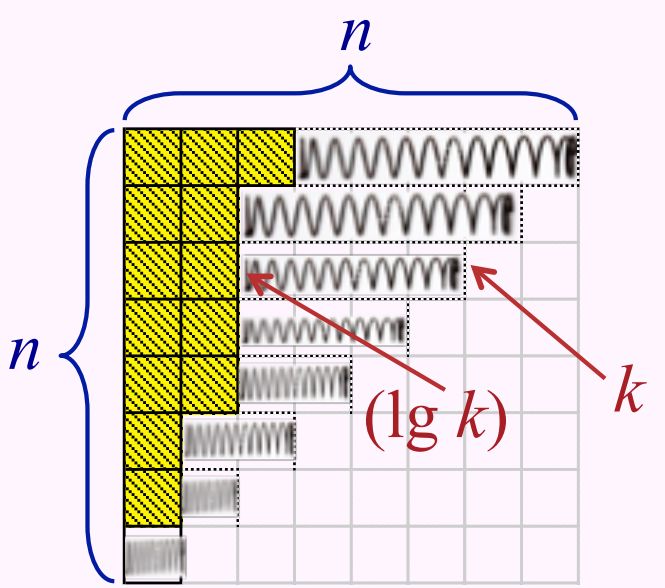
Sum-Log Pattern

AA-Pattern: Sum-Log

$$T(n) = \sum_{k=1}^{n} (\lg k) = \Theta(n \lg n)$$

$$= (\lg 1) + (\lg 2) + \ldots + (\lg n)$$



$T(n) = \Theta(n \lg n)$

Code Pattern

SUM-LOG (A, n, X) ▷ $A[1..n]$

(* bin-search $A[1..k]$, k=1,2,…,n *)

for $k \leftarrow n$ **downto** 1

Binary-Search$(A, 1, k, X)$

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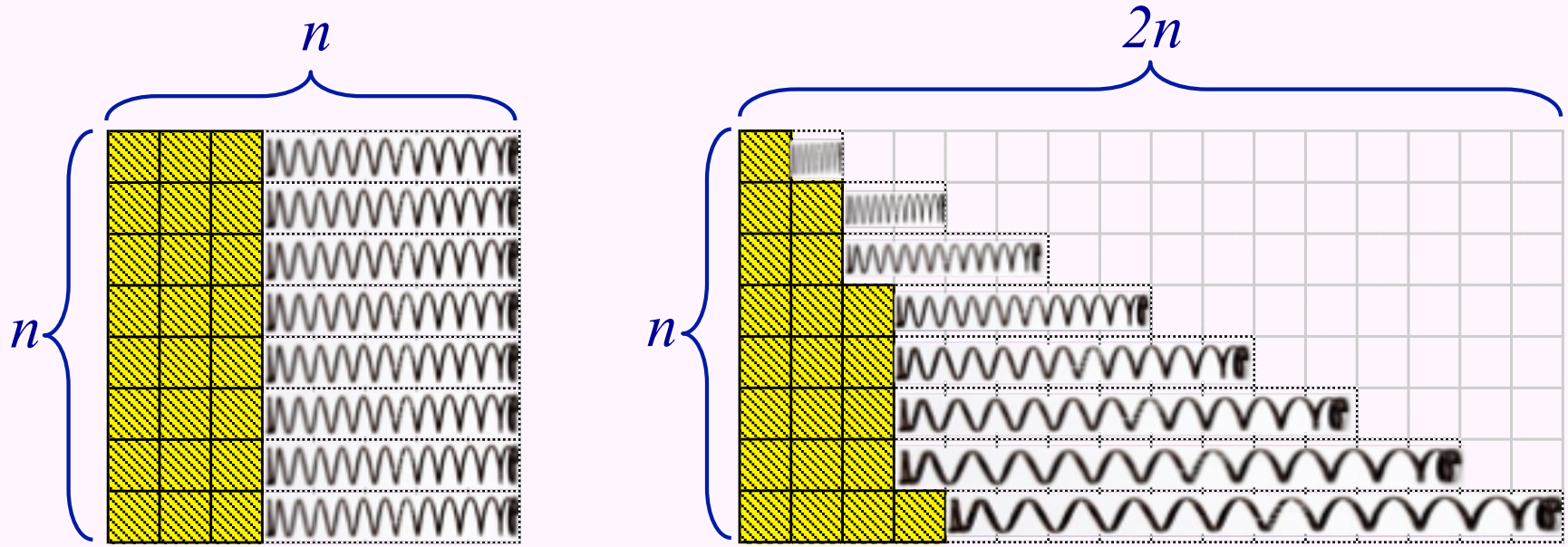
Sum-Log Pattern (variants)

AA-Pattern: Sum-Log

$$T(n) = \sum_{k=1}^{n} (\lg k) = \Theta(n \lg n)$$
$$= (\lg 1) + (\lg 2) + \ldots + (\lg n)$$



$T(n) = \Theta(n \lg n)$ $\qquad$ $T(n) = \Theta(n \lg n)$

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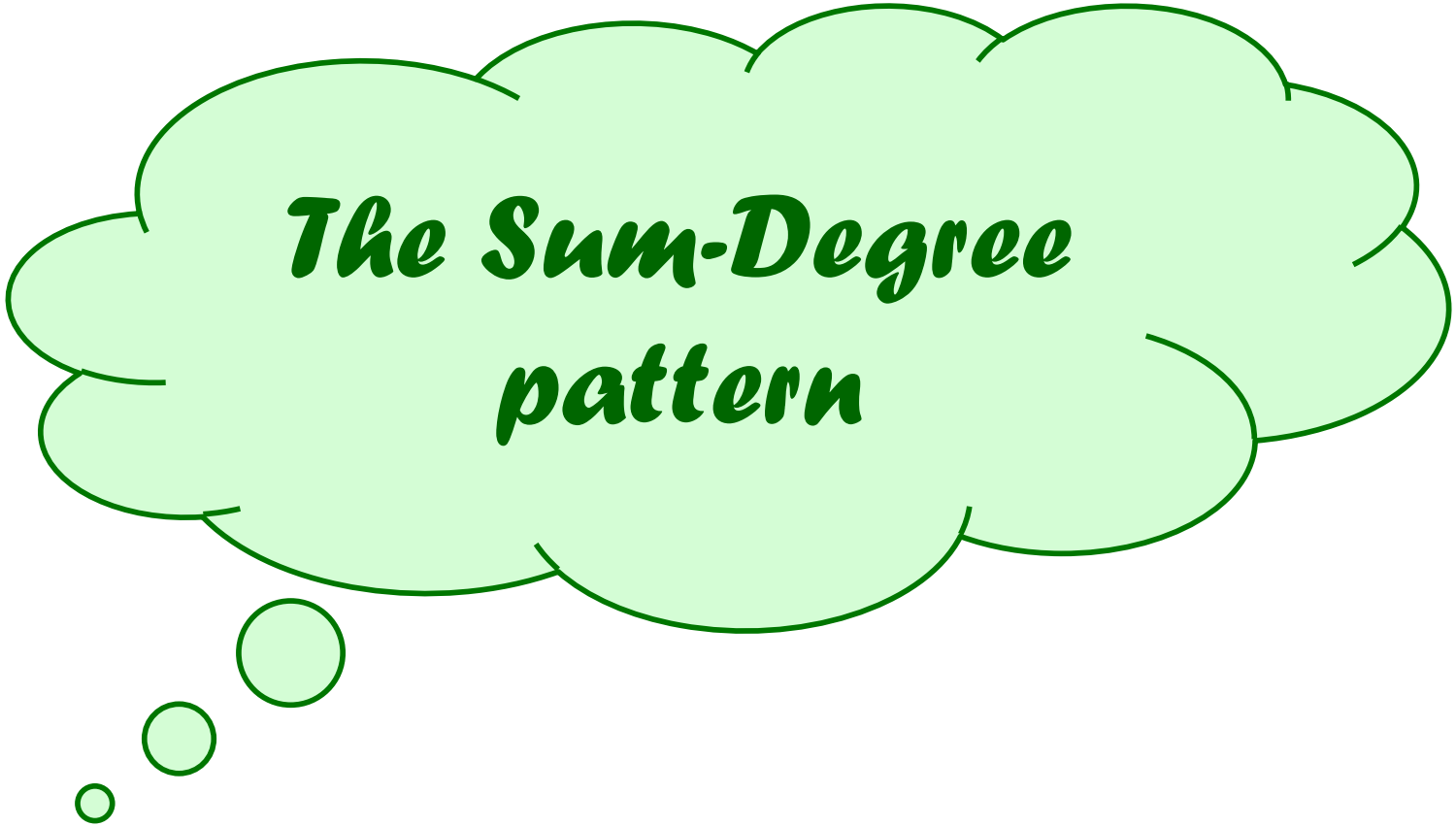
Algorithm-Analysis Pattern?



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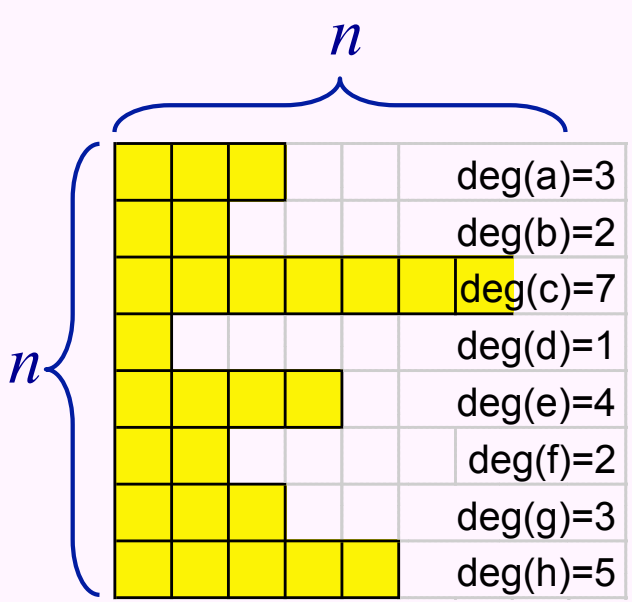
Sum-Degree Pattern

AA-Pattern: Sum-Degree

$$T(n,m) = \sum_{k=1}^{n} \deg(v_k) = 2m = \Theta(n+m)$$

$$= \deg(v_1) + \deg(v_2) + \ldots + \deg(v_n)$$



$T(n,m) = \Theta(n+m)$

Code Pattern

SUM-DEGREE (G, n, m) ▷ $G=(V,E)$

(* *V has n nodes, E has m edges* *)

for each *v in G* **do**

for each *w in Adj(v)*

Print details of (v, w)

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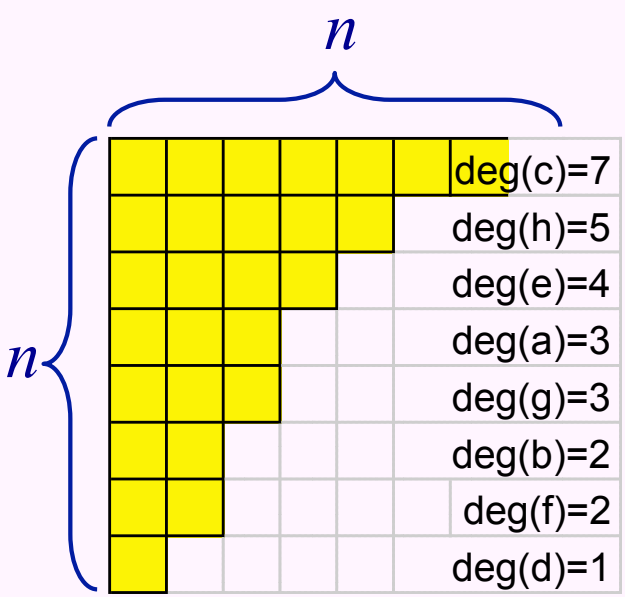
Sum-Degree Pattern (variants)

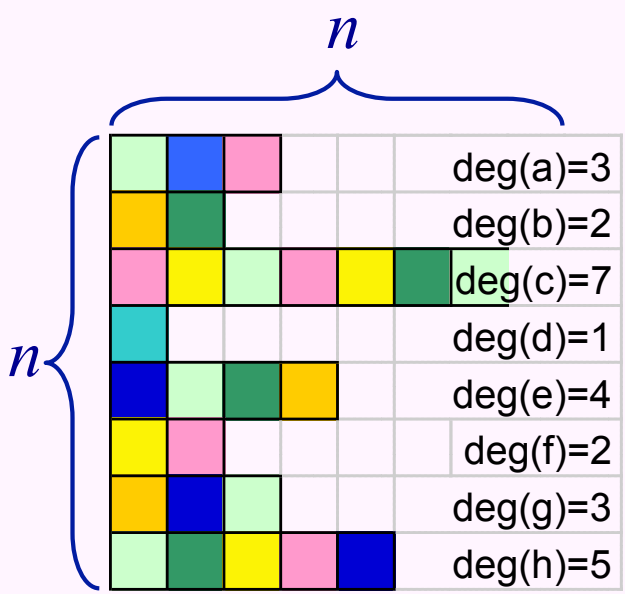
AA-Pattern: Sum-Degree

$$T(n,m) = \sum_{k=1}^{n} \deg(v_k) = 2m = \Theta(m)$$

$$= \deg(v_1) + \deg(v_2) + \ldots + \deg(v_n)$$



$T(n) = \Theta(n+m)$



$T(n) = \Theta(n+m)$

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HW for myself and students…

This is for CS3230 staff and students:

Anyone interested in volunteering to help in this, please email me and say you like to help.

Thanks in advance.

- For each AA-pattern, list the different algorithms that has that AA-pattern.
- Do it for all the algorithms covered in CS1020, CS2010, CS2020.
- Discover new AA-patterns

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HW for myself and students…

- **IDEA by HW:**
  - Create flash-card of each AA-pattern.
  - From page has definition of pattern, back page has variant pattern and corresponding algorithms;

Looking for people who are good with artistic design to prototype some
nice, cool, AA-pattern flash-cards

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Thank you.

Q & A

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